Sentimental Analysis Of Twitter Data

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***Abstract*—Analyzing sentiment from Twitter data is crucial for grasping public perception in areas like politics, healthcare, and finance. Nonetheless the casual and disruptive character of Tweets that contain brevity, informal language, and sarcasm complicate accurate sentiment analysis. This document showcases a framework for sentiment analysis that assesses conventional machine educational frameworks, advanced deep learning methods, combined strategies, and a specialized transformer model based on Twitter data. Conventional models such as Logistic Regression, Multinomial Naive Bayes and Support Vector Machine were utilized, succeeded by a Long Short-Term Memory network deeplearning model. Two hybrids models TF-IDF combined with SVM and Word2Vec combined with LSTM enhanced performance attaining accuracies of 88.6 and 89.3 respectively. To boost performance even more, a BERTweet model that was pre trained on extensive Twitter data was refined for three class sentiment analysis and reached around 91 accuracy. Interpretable AI employing SHAP was established to offer interpretability at the token level. Findings demonstrate the efficacy of transformer-based sentiment examination.**

***Index Terms*—Sentiment Analysis, Twitter Data, BERTweet, Hybrid Models, Explainable Artificial Intelligence, SHAP, Natural Language Processing, Transformer Models**

I. INTRODUCTION

Social media sites have emerged as a key channel for conveying thoughts, feelings, and responses to reality happenings. Among these platforms Twitter stands out as a good source of real time textual information and is extensively utilized for analysis in various fields including such as politics, healthcare,economics, finance, and social issues. Despite its usefulness sentiment analysis on Twitter continues to be difficult because of the brevity, casual and loud characteristics of tweets, which frequently contain slang, emojis, abbreviations, and irony creating precision interpretation challenging.

Initial research on Twitter sentiment analysis predominantly depended on lexicon based approaches and conventional machine learning algorithms. Even though these methods are computationally efficient these approaches frequently do not convey contextual meaning and semantic connections inherent in social media text. Due to the presence of bigger Twitter datasets utilized deep learning models such as Long Short-Term Memory networks were introduced to model sequential patterns.Although models based on LSTM have enhanced effectiveness compared to conventional methods they remained limited in handling long range context and ambiguous expressions commonly found in tweets.

To improve these limitations hybrid approaches combining feature based representations with learning-based models have been explored.TF-IDF with Support Vector Machine (SVM) and Word2Vec with LSTM are evaluated to balance computational efficiency and representation learning. Although these hybrid models demonstrate improved performance compared to baseline methods, they are still insufficient for capturing complex contextual semantics.

Recent advances in transformer based architectures have further improved sentiment analysis through self-attention mechanisms. In particular a transformer model BERTweet pre trained on Twitter data exhibits strong capability in learning tweet specific linguistic patterns. However despite its high predictive accuracy the model lacks interpretability. To address this issue, a unified Twitter sentiment analysis framework is proposed that evaluates traditional, hybrid, and transformer-based models while integrating Explainable Artificial Intelligence (XAI) using SHAP to provide transparent, token-level explanations of model predictions.

To give a summary of the suggested methodology, Fig. 1 illustrates the workflow of the Twitter sentiment analysis framework used in this study. The figure summarizes the key stages, including data preprocessing, feature extraction, baseline and hybrid modeling, BERTweet fine-tuning, sentiment prediction, and explainability using SHAP, offering a clear view of the overall approach.

II. LITERATURE REVIEW

The analysis of sentiments expressed on Twitter has grown to be a significant subject in natural language processing due to this platform mirrors public sentiment instantly and on a broad scale. Unlike traditional sentiment analysis on long documents Twitter data is short, informal, and highly dynamic. Tweets often include emojis, hashtags, slang, sarcasm, and rapidly changing discussion topics, which makes sentiment interpretation difficult and calls for specialized modeling techniques.The sentiment analysis has been widely studied due to its impact on public opinion, finance, healthcare, and political discourse. Early research mainly relied on lexicon-based and traditional machine learning methods. Qi and Shabrina [1] showed that such approaches struggle to capture contextual meaning in short and noisy tweets, while Hassan et al. [2] demonstrated that emotion-based sentiment analysis can explain cryptocurrency market trends but remains limited to specific domains.

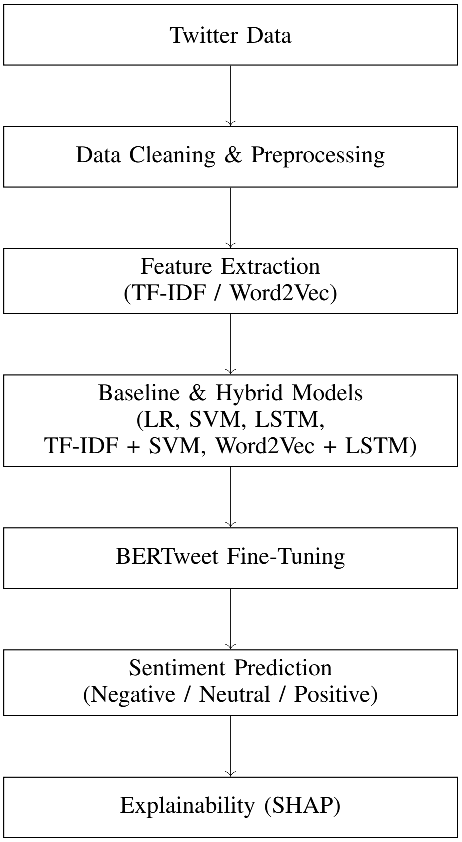

Fig. 1. Proposed Twitter sentiment analysis workflow.

TABLE I
SUMMARY OF TWITTER SENTIMENT ANALYSIS STUDIES

Ref.	Method	Domain / Dataset	Key Limitation
[1]	Lexicon + ML	General Twitter	Weak contextual understanding
[2]	Emotion Lexicon	Cryptocurrency Tweets	Domain-dependent insights
[3]	DLMNN + Hadoop	Large-scale Twitter	High computational cost
[4]	ML + VADER	Election Tweets	Demographic bias
[5]	ULMFiT + SVM	Airline / Debate Tweets	Poor interpretability
[7]	Ensemble GCR-NN	Racism-related Tweets	Task-specific model
[9]	BERT-based DL	Multi-domain Twitter	Black-box behavior
[10]	ML + TF-IDF	Monkeypox Tweets	Manual labeling reliance
[14]	RoBERTa-LSTM	Sentiment140, IMDb	Limited explainability
[16]	RoBERTa-GRU	Airline Tweets	Increased model complexity
[17]	Lexicon + BERT	Crime-related Tweets	Sensitive to lexicon quality
[18]	XLM-T Transformer	Multilingual Twitter	Heavy pretraining required
[15]	Transformer Ensemble	Cross-lingual Tweets	Translation errors
[20]	Survey / Review	Multi-domain	No empirical validation

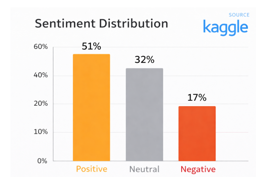

Fig. 2. Sentiment distribution of the Kaggle Twitter dataset.

With the rapid growth of Twitter data, deep learning models became increasingly popular. Neelakandan et al. [3] proposed a Deep Learning Modified Neural Network (DLMNN) supported by large-scale preprocessing to improve sentiment classification accuracy.

Recent studies have focused on transformer-based and hybrid architectures to enhance contextual understanding. Al-Badani et al. [5] combined ULMFiT with SVM to outperform traditional classifiers, while Bello et al. [9] demonstrated that BERT-based models significantly outperform Word2Vec-based methods. Hybrid models such as RoBERTa-LSTM and RoBERTa-GRU further improved performance by addressing long-range dependencies and class imbalance [14], [16].

Domain-specific applications have also been explored. Bengesi et al. [10] analyzed public sentiment during the Monkeypox outbreak, and Lee et al. [7] achieved high accuracy in detecting racist content using ensemble deep learning models. Boukabous and Azizi [17] further showed that combining lexicon-based labeling with BERT is effective for crime-related sentiment analysis.

Multilingual sentiment analysis remains an active research area. Garg and Sharma [11] emphasized robust preprocessing for multilingual tweets, while Barbieri et al. [18] introduced XLM-T, a Twitter-specific multilingual transformer. More recently, Miah et al. [15] demonstrated that ensemble-based cross-lingual sentiment analysis using translation is effective for low-resource languages.

Despite these advances, most existing approaches prioritize accuracy while offering limited interpretability. Challenges related to bias, domain generalization, and explainability remain open, motivating the development of explainable and domain-aware transformer-based sentiment analysis frameworks [20].

Table I provides a structured summary of representative studies on Twitter sentiment analysis, outlining the methodologies adopted, the application domains or datasets examined, and the principal limitations observed across traditional, deep learning, and transformer-based approaches.

III. DATASET AND DATA PROCESSING

A. Dataset Description

The dataset utilized in this research was obtained from Kaggle and comprises of Twitter posts labeled for sentiment analysis. Each tweet is labeled into one of three sentiment categories: positive, neutral, or negative. Twitter was chosen as the data source due to its real time nature. The dataset captures the characteristics of social media text including short sentence length, informal language, hashtags, user mentions, emojis, abbreviations, and domain-specific expressions. These properties introduce significant variability and noise, making the dataset both challenging and appropriate for evaluating traditional, hybrid, and transformer-based sentiment analysis models.

The overall distribution of sentiment classes in the dataset is illustrated in Fig. 2, which shows that positive tweets form the majority, followed by neutral and negative samples.

B. Data Cleaning and Preprocessing

Before the model training the unprocessed Twitter data went through multiple preprocessing steps to minimize noise and enhance overall data quality. These steps included the removal of URLs, user tags, hashtag symbols, numerical values, special characters, and extra whitespace. All text was converted to lowercase to keep the data consistent. Each tweet was divided into individual tokens using tokenization, and then removal of stopwords to remove significantly frequent but semantically neutral words. These preprocessing functions are useful in the reduction of data sparsity and promote the efficiency of the following feature extraction and model learning.

The data preprocessing workflow is illustrated in Fig. 3. It breaks down the major stages involved in noise elimination, tokenization, normalization, and emoji handling, resulting in clean, normalized and filtered tweets suitable for feature extraction and sentiment classification.

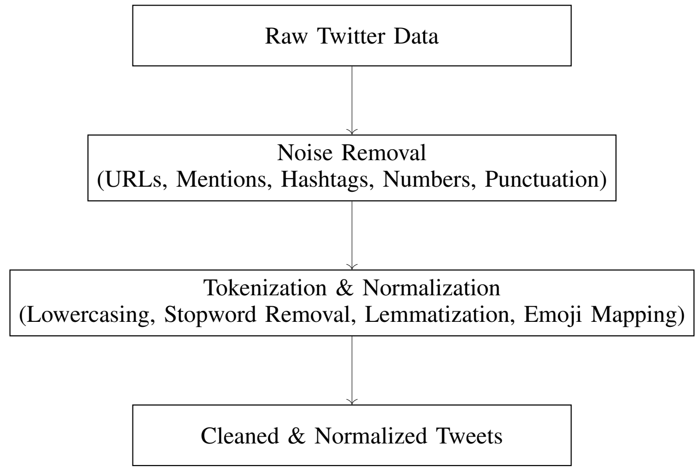

Fig. 3. Data preprocessing pipeline for Twitter sentiment analysis.

C. Feature Extraction

There are two feature extraction techniques that were used in the case of the baseline and hybrid models. TF-IDFT is important because it was distributed across the frequency distribution to understand corpus, while Word2Vec embeddings were used for meaningful associations between words in a continuous vector space. TF-IDF features were mostly used with conventional machine learning models whereas Word2Vec embeddings were combined with deep learning models to incorporate contextual information.Transformer-based models such as BERTweet do not require the explicit feature extraction, as such they learn the contextual representations directly out of raw text.

D. Preparation of Data for Modeling

Upon cleaning the data and feature extraction, the data have been extracted, split that data into training and testing sets to see how well the model performs. Partitions were also constant between data via baseline, hybrid as well as for transformer based models to ensure fair and sound comparison. This consistent data processing pipeline improves the reliability of the results and makes the experiments easier to reproduce.

IV. Methodology

This describes the sentiment analysis strategy used in this study. It encompasses classical models of machine learning, hybrid approaches, deep learning strategies,transformer model and explainability features and based model. The overall workflow is illustrated in Fig. 1.

A. Traditional Machine Learning Models

The traditional machine learning models were considered to form the baseline classifiers due to their simplicity, computational complexity and interpretation. In order to enable such models to process textual data, tweets first were changed into numerical forms using Term Frequency Inverse Document frequency which assigns more weight to words should they be frequency repeated in the same tweet but not excessively repeated in all tweets. The TF-IDF value is calculated as

$$\text{value}(t, d) = \text{frequency}(t, d) \times \log\left(\frac{n}{n_t}\right) \qquad (1)$$

where t denotes a term, d represents a tweet (document), $\text{frequency}(t, d)$ is the frequency of term t in tweet d, n is the total number of tweets in the dataset, and n_t is the number of tweets containing the term t. The TF-IDF formulation in Eq. (1) follows the standard definition reported in [21] and tweets are good with it since they just highlight the words that are actually important and deemphasize the common words that appear everywhere. This enhances easier interpretation of what a tweet means.

Three classical machine learning classifiers- Linear Regression, Multinomial Naive bayes, Support Vector Machine were tested as a reference model to sentiment analysis based on TF-IDF features representations. These are usually used in the process of classifying text and are reliable as a standard of evaluating more complex methods.

A probabilistic classifier that was used inLinear Regression was to understand the relationship between TF-IDF features and sentiment labels.The model assigns weights to individual features that estimates the likelihood of a tweet being contained in one sentiment category. Its straightforwardness and effectiveness establish it as a strong base of high dimensional text information.

Support Vector Machine was employed to create an perfect decision boundary among sentiment categories within the feature space. SVM is good in terms of max performance reducing the distance between classes particularly handling with sparse representations like TF-IDF vectors.

Multinomial Naive Bayes was applied as a generative model that shows distribution of words of classes depending on the condition of independent features. Even with its simplicity this model remains computationally efficient, and has no assumptions often generates competitive results in sentiment analysis tasks.

These traditional classifiers used together establish baseline performance metrics, and it can be compared to hybrid and deep learning and transformer models discussed later in the study significantly.

B. Hybrid Models

To increase the representational ability and retain Two hybrid models have been investigated, which are computational efficienct.

1) Hybrid Model 1: TF-IDF + SVM: In the first hybrid approach, combination of TF-IDF features and an SVM was used to improve discrimination between sentiment classes. This model has the advantage of making use of statistically weighted features and margin based classification, which is more effective than traditional models on their own.

2) Hybrid Model 2: Word2Vec + LSTM: The second hybrid approach merges Word2Vec embeddings with a LSTM network to understand both semantic temporal and causal relationships on tweets. Word2Vec encodes words as dense vectors that maintain semantic associations, which are processed in sequence by the LSTM. The LSTM recognizes contextual and sentiment trends within the tweet Through the use of an internal memory that are frequently overlooked by feature based methods. This hybrid approach provides finer representation of sentiment as compared to conventional classifiers and at the same time computationally more more efficient than transformer based models.

C. Deep Learning Model

A separate LSTM-based deep learning system was also implemented directly to model the sequence of tweets representations. LSTM networks are capable of learning time-dependence and their ability to handle long-range context information in tweets remains limited, motivating the use of model on the basis of transformers.

D. Transformer-Based Model: BERTweet

In order to gain more contextual insight, BERTweet, a model trained on large-scale Twitter based on transformers data was refined in sentiment classification. BERTweet is based on the transformer architecture founded on representational models to express contextual relationships: self-attention mechanisms The model is built on the transformer architecture and employs self-attention mechanisms to learn contextual dependencies between words in a sequence, enabling more accurate sentiment prediction [22], [25]. The self-attention operation employed by the transformer is defined as

$$\text{Attention}(X, Y, Z) = \text{softmax}\left(\frac{XY^T}{\sqrt{D_k}}\right) Z \tag{2}$$

where X, Y, and Z denote the query, key, and value matrices, respectively, and D_k represents the dimensional of the key vectors. Equation (2) follows the standard transformer self-attention formulation [22] and adopted by BERTweet for contextual representation learning [25]. This mechanism helps the model understand important relationships between words across the tweet.

TABLE II
COMPARISON OF SENTIMENT ANALYSIS MODEL PERFORMANCE

Model	Accuracy	Precision	Recall
Linear Regression	84.2	83.6	82.9
Naïve Bayes	82.5	81.9	81.2
SVM	86.1	85.4	84.8
TF-IDF + SVM	88.6	87.9	87.2
Word2Vec + LSTM	89.3	88.7	88.1
LSTM	87.4	86.8	86.1
BERTweet	91.57	90.4	90.1

The final sentiment prediction is obtained using a softmax classifier Equation (3) , which converts the model output scores into normalized class probabilities for multi-class sentiment classification [23]. The softmax-based probability estimation is given by

$$P(y = c|x) = \frac{exp(s_c)}{\sum_{j=1}^{C} exp(s_j)} \tag{3}$$

where C denotes the total number of sentiment classes and s_c is the predicted score for the class c.

E. Explainable Artificial Intelligence Using SHAP

To enhance model transparency, Explainable Artificial Intelligence (XAI) was incorporated using SHAP , which interprets predictions through additive feature attributions derived from cooperative game theory [24]. The SHAP explanation model is defined as

$$g(x) = \phi_0 + \sum_{i=1}^{n} \phi_i \tag{4}$$

where $g(x)$ denotes the model output, ϕ_0 represents the base value, and ϕ_i indicates the contribution of the i^{th} token. This formulation enables token-level interpretation of sentiment predictions and mitigates the black-box nature of transformer-based models [24].

Fig. 4 illustrates the BERTweet-based sentiment analysis workflow, highlighting the major stages from preprocessing to explainable sentiment prediction.

V. RESULTS, EVALUATION, AND ERROR ANALYSIS

This part covers discusses the experimental results obtained from conventional machine learning models, hybrid approaches, neural network models, along with a transformer-oriented sentiment analyzer. Model performance is assessed through standard classification metrics to allow for an equitable and uniform comparison throughout all methods.

A. Evaluation Metrics

The performance of the sentiment classification models was measured using three standard metrics: Accuracy, Precision, and Recall, which are frequently employed in supervised classification assignments [23]. Precision assesses the overall

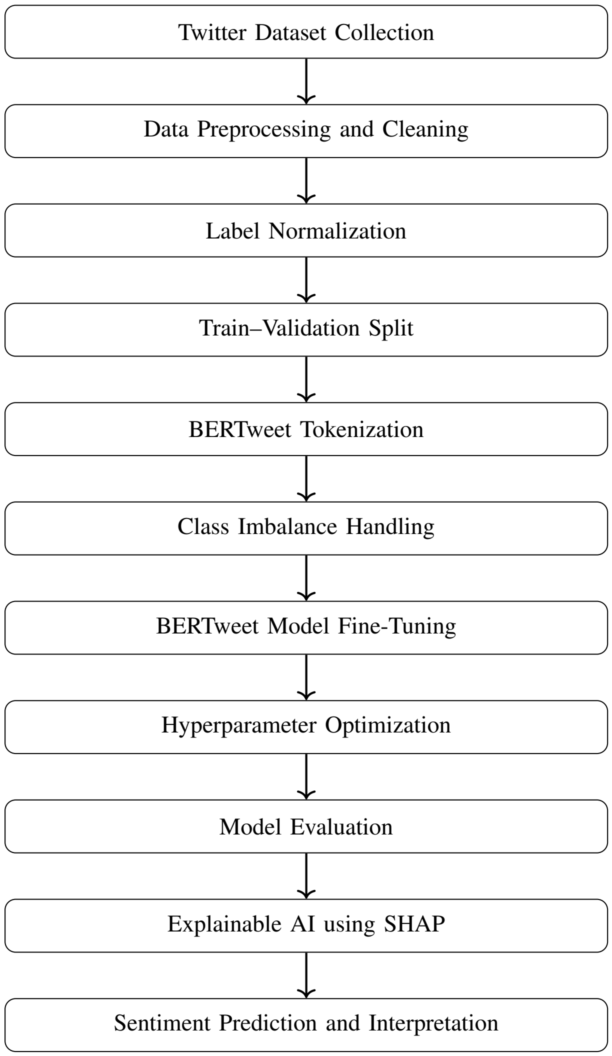

Fig. 4. BERTweet-based sentiment analysis workflow with explainability.

accuracy of forecasts, Precision measures the ratio of accurately forecasted positive cases, while Recall assesses the capability of the model to recognize all pertinent positive instances.These metrics are defined as follows:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \tag{5}$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{6}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{7}$$

B. Model Performance Comparison

TTable II outlines the performance of every assessed models regarding accuracy, precision, and recall. Among among these, the BERTweet model achieves the highest accuracy of 91.57

The results show that traditional machine learning models provide a strong baseline for sentiment classification. Hybrid

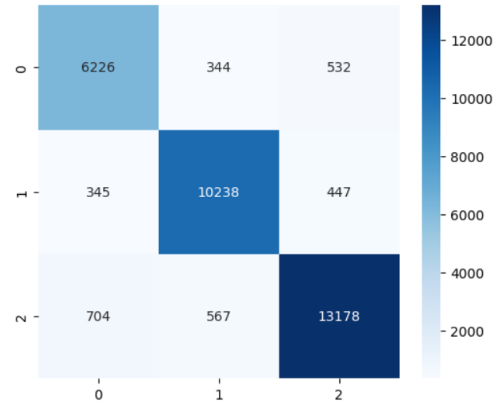

Fig. 5. Confusion matrix of the BERTweet model for three-class sentiment classification.

models further enhance performance by combining feature-based representations with learning-based techniques. Among all assessed methods, the transformer-driven BERTweet. The model reaches the top scores in every metric, showcasing its capability to accurately grasp contextual and semantic nuances data in Twitter information.

C. Error Analysis

Although the overall performance is strong, several sources of classification errors were identified. Traditional and hybrid models often misclassified tweets containing sarcasm, idiomatic expressions, or sentiment that depends heavily on context. Neutral tweets were particularly difficult to classify due to lexical overlap with both positive and negative classes. While the BERTweet model significantly reduced such errors, misclassifications still occurred in tweets with implicit sentiment or ambiguous phrasing. The use of SHAP-based explainability allowed marking the influential tokens accountable for misleading forecasts, offer excellent guidance to the model conduct and point out possible trend for future improvement. Fig. 5 illustrates the confusion matrix of the BERTweet model, where rows represent the real sentiment labels and columns suggests predicted labels, with 0, 1, and 2 representing negative, neutral, and positive classes, respectively.

Discussion

The results of the conducted experiment indicate a clear difference in performance comparisons between traditional, hybrid and transformer-based methodologies of sentiment analysis with an emphasis on the complexities. Twitter text and the merits and demerits of both modeling technique. The traditional machine learning models like Linear Regression, Naive Bayes, and SVM are useful but depend on manually and are limited by their reliance on manually constructed features, which restrict their ability to capture contextual and implicit sentiment that is generally found in tweets. Hybrid models show continuous improvements per- representations with feature-based representations using combinations, performance can be combined learning-driven methods. TF-IDF + SVM and Word2Vec + LSTM obtain more balanced and more accurate comparative measurements of precision-recall to other traditional methods, pointing out the benefits of incorporating semantic data. However, these gains are small and this implies that hybrid models are still struggling with long-range context.

Transformer-based model Bertweet performed best in terms of overall performance on all evaluation metrics since the model is pre-trained on huge amounts of Twitter data. Nevertheless, despite the high level of accuracy, the model continues to fail in its attempt to identify the neutral sentiment correctly, which is reflective of the inherent ambiguity of the social media language. The use of SHAP-based explainability as an additional means of strengthening the method also makes the sentiment analysis system more transparent and reliable as it shows the words that impact predictions.

Conclusion and Future Work

This study examined the Twitter sentiment analysis using a systematic comparison of the classical machine learning models, combinations and a transformer architecture. As can be seen, the results reveal a gradual performance improvement the models shift to less complex lexical representations context aware methods of learning.

Classifiers such as Linear regression, Naive Bayes and SVM are reliable as a baseline. They are however reliant on handcrafted features not only limit their ability to address the informal, situational nature of Twitter text. The hybrid models offer a sensible trade-off between representation and efficiency learning. The TF-IDF + SVM and Word2Vec + LSTM models provide more consistent and has better performance than conventional algorithms especially in high-noise data environments. These gains not withstanding, the improvements are moderate that show that hybrid models are yet to be fully realized challenge in the representation of long range dependencies and subtlety. sentiment changes that are common in twitters.

The BERTweet model based on a transformer. architecture, overall attains the highest performance, with a maximum accuracy of 91.0. This is because of its effectiveness. to pretraining on Twitter data and its capability to pretrain on large scale data. contextual relationships based on self attention. However There are difficulties in using complex models transformers would bring interpretability. This shortcoming is overcome by use of SHAP based explainability that gives out explicit token level understandings of behavior on predictions.

On the whole the proposed framework creates an equilibrium between Making it well predictive performance and interpretability appropriate to real world sentiment analysis. Future work are able to investigate more successful approaches to neutral sentiment management class imbalance as well as extrapolating the framework to multilingual and cross-domain social media data.

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